

Schistosomiasis Endgame Costing Tool

Technical Manual and User Guide

Tool version: 1.3.0

Draft for production release - July 13, 2026

Audience: national NTD programme managers, health economists, implementation analysts, and technical maintainers.

Purpose. This manual explains how to install, configure, run, and interpret the Schistosomiasis Endgame Costing Tool. It also documents the input data structure, analysis workflow, outputs, assumptions, and troubleshooting procedures needed before production use.

Document control

Item	Detail
Tool name	Schistosomiasis Endgame Costing Tool
Version documented	1.3.0
Manual status	Draft for production release and user acceptance testing
Primary runtime	Streamlit web app backed by Python modules in schisto_tool/
Required production datasets	datasets/consolidated_schisto.csv and datasets/df_gdp.csv
Primary users	Programme managers, health economists, policymakers, and technical analysts
Primary outputs	Burden estimates, DALYs, ICERs, CEACs, productivity gains, health-sector costs, ROI, economic impact, budget impact, sensitivity analysis, and elimination projections

Revision history

Manual version	Date	Description
0.1	2026-07-13	Initial technical manual drafted from the production bundle and QA notes.
0.3	2026-07-13	Prevalence consistency update: unified biennial annual-equivalent effect across PSA, sensitivity, and trajectory views; added species-specific trajectory decay defaults and discrete no-MDA compounding.

Table of contents

- 1. Overview
- 2. Quick start for analysts
- 3. Installation and deployment
- 4. Data requirements
- 5. Step-by-step tool workflow
- 6. Interpreting results and downloads
- 7. Technical assumptions and formulas
- 8. Sensitivity and elimination analysis
- 9. Quality assurance before production
- 10. Troubleshooting
- 11. Appendices: data dictionaries, output files, and glossary

1. Overview

The Schistosomiasis Endgame Costing Tool is a Streamlit-based economic and budget analysis application for schistosomiasis mass drug administration (MDA) programmes. It uses country economic inputs and ESPEN-style schistosomiasis surveillance data to generate estimates by country, administrative unit, disease module, and intervention scenario.

The tool can be used for national-level planning, subnational prioritization, scenario testing, investment case preparation, and budget discussions. It is not a clinical decision tool and should be interpreted as a model-based planning tool.

1.1 What the tool estimates

- **Disease burden:** infected population, morbidity-state caseloads, and annual DALYs with and without MDA.
- **Programme costs:** drug, delivery, mapping/M&E, training, supervision, other annual programme costs, and biennial off-year fixed costs.
- **Cost-effectiveness:** DALYs averted, incremental cost, ICER, net monetary benefit, CEAC, and probability cost-effective at country-specific thresholds.
- **Economic impact:** programme costs, health-sector savings, productivity gains, cumulative net economic benefit, and benefit-cost ratio.
- **Budget impact:** gross payer costs, clinical-management offsets, staff time saved, and cumulative net budget impact.
- **Endgame projections:** prevalence trajectories and probability of reaching elimination as a public health problem under selected coverage and frequency assumptions.

1.2 Disease modules

Module	Use when	Core morbidity states
S. mansoni (intestinal)	The selected geography has intestinal schistosomiasis or the user wants an intestinal/hepatosplenic burden scenario.	Anemia, hepatomegaly, periportal fibrosis, portal hypertension, esophageal varices.
S. haematobium (urogenital)	The selected geography has urogenital schistosomiasis or the user wants a urinary/FGS/cancer burden scenario.	Hematuria, hydronephrosis, female genital schistosomiasis, S. haematobium-attributable bladder cancer.
Both species	The geography has evidence for both species or the programme wants a combined investment case.	Runs both modules and calculates combined DALYs, benefits, programme ICER, and ROI.

1.3 User roles

Role	Typical tasks
Programme analyst	Select country/geography, set MDA assumptions, run PSA and sensitivity analysis, export result tables.
Health economist	Review costs, economic inputs, DALYs, ICERs, NMB, ROI, economic impact, and budget impact.
Data manager	Prepare and validate ESPEN and country economic input files.
Technical maintainer	Deploy app, manage requirements, run tests, troubleshoot runtime/data issues, and maintain version control.

2. Quick start for analysts

Use this section when the application has already been deployed and the production datasets have already been installed.

1. Open the Streamlit application in a browser.
2. In the left sidebar, select a country.
3. Choose the disease module: *S. mansoni*, *S. haematobium*, or Both species.
4. Choose the geographical unit: National level, ADMIN1, ADMIN2, or implementing unit.
5. Review the MDA programme parameters. Coverage defaults to the most recent PopTreat / PopReq ratio for the selected geography but can be changed.
6. Review programme costs, country economic parameters, and budget-impact resource inputs.
7. Open the Disease inputs tab and verify the default prevalence and population values for each selected species.
8. Run or refresh the PSA if auto-run is off, or if you changed inputs and want to force a refresh.
9. Review the Results, Economic impact, Budget impact, and Elimination Projections tabs.
10. Download CSV outputs needed for reporting, quality assurance, or further analysis.

2.1 Minimum analyst checklist

- ☐ Country and geographic unit match the planning question.
- ☐ Disease module matches the species present in the selected unit.
- ☐ PopReq, PopTreat, SAC prevalence, and adult prevalence look plausible in the Country inputs tab.
- ☐ Target population multiplier reflects whether adults are included.
- ☐ Cost inputs reflect the scenario being presented.
- ☐ PSA iterations and seed are recorded for reproducibility.
- ☐ ICER and ROI are interpreted from the combined programme output when both species are selected.
- ☐ Budget impact is not mixed with productivity gains; economic impact is the section that includes productivity gains.

3. Installation and deployment

3.1 Expected production package

The production package should use normalized file names and a Python package layout. Do not deploy uploaded working-copy names such as app(4).py or config(3).py.

```
production_bundle/
  app.py
  schistosomiasis_data_uploader.py
  requirements.txt
  datasets/
    consolidated_schisto.csv
    df_gdp.csv
  schisto_tool/
    __init__.py
    cache.py
```

```

caseloads.py
config.py
data.py
economics.py
elimination.py
health_sector.py
parameters.py
prevalence.py
sensitivity.py
simulation.py
ui_helpers.py
utils.py
visualization.py
tests/
  smoke_core.py
  test_regressions.py

```

3.2 Install dependencies

Create an isolated Python environment and install the production requirements.

```

cd production_bundle
python -m venv .venv
# macOS/Linux
source .venv/bin/activate
# Windows PowerShell
# .venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements.txt

```

The production requirements file includes Streamlit, Altair, NumPy, pandas, SciPy, and OpenPyXL. OpenPyXL is required for Excel input handling in the ESPEN uploader.

3.3 Add production datasets

Place the following files under the datasets/ directory before launching the app:

- datasets/consolidated_schisto.csv - ESPEN-style schistosomiasis data or a prepared canonical dataset.
- datasets/df_gdp.csv - country economic inputs used for wages, GDP-based thresholds, life expectancy, inflation, and OPD/IPD unit costs.

3.4 Launch the app

```

cd production_bundle
streamlit run app.py

```

After the app starts, the country list should populate in the sidebar. If the app stops with a missing dataset message, go to Section 10.

3.5 Recommended deployment checks

```

python -m py_compile app.py schistosomiasis_data_uploader.py schisto_tool/*.py
PYTHONPATH=. python tests/smoke_core.py
PYTHONPATH=. pytest -q tests/test_regressions.py

```

Run these checks after dependency installation and before production sign-off. They confirm syntax, core computational imports, smoke-level model execution, and regression coverage for key production blockers.

4. Data requirements

4.1 ESPEN schistosomiasis dataset

The app reads datasets/consolidated_schisto.csv and processes it through the ESPEN loader. The loader can normalize common column aliases, but the prepared dataframe used by the app has the canonical columns below.

Column	Meaning	Scale / notes
ADMIN0	Country name	Must match Country in df_gdp.csv.
ADMIN1	First administrative unit	Used for geographic filtering.
ADMIN2	Second administrative unit	Used for geographic filtering.
IUs_NAME	Implementing unit name	Lowest geographic selector in the app.
PopReq	Population requiring MDA	Main denominator for programme and burden defaults.
PopTreat	Population treated in last recorded MDA data	Used to calculate default MDA coverage.
Prev_SAC	School-age children prevalence	Percentage points, 0-100.
Prev_Adults	Adult prevalence	Percentage points, 0-100. If missing, the loader can fall back to SAC prevalence.
Sch_MDA_Rounds	Recorded schistosomiasis MDA rounds	Maximum value is shown for the selected unit.
sm_prev_pct	S. mansoni prevalence	Percentage points, 0-100.
sh_prev_pct	S. haematobium prevalence	Percentage points, 0-100.
sm_share, sh_share	Species shares at row level	Used for species-specific allocation where available.
sm_share_pct, sh_share_pct	Country-level species share defaults	Used to set the initial mixed-species slider value.
species, species_source	Inferred species classification and evidence source	Useful for QA and troubleshooting.

4.2 Country economic input file

The app reads datasets/df_gdp.csv. At minimum, it must contain the columns used by the sidebar economic inputs and country selector.

Column	Used for
Country	Country selector and join key to ADMIN0.
Annual_PPP(Int\$)	Per capita GDP PPP, GDP-based threshold, daily wage calculation.
inequality_.2 Quintile	Bottom-quintile income share used for inequality-adjusted wage.
Weekly_Work_Hours	Hourly and daily wage calculation.
Life_Expectancy	Years of life lost calculation for bladder cancer.
Primary Hospital_OPD_Costs	Base-year OPD unit cost.
Primary Hospital_IPD_Costs	Base-year IPD unit cost.
Inflation rate (consumer prices) (%)	Medical inflation adjustment for current OPD/IPD costs.
Hourly_PPP(Int\$)	Optional. If missing, it is computed from Annual_PPP(Int\$) and Weekly_Work_Hours.

4.3 Preparing ESPEN data from raw files

The uploader can read CSV or Excel files, normalize geographic and prevalence aliases, infer species, optionally consolidate duplicate geographies, and save a cleaned file for use by the app.

```
python schistosomiasis_data_uploader.py raw_espen_file.csv datasets/consolidated_schisto.csv --head 10
```

Common options include `--prevalence-scale auto|percent|fraction`, `--unknown-species-strategy both|none|mansoni|haematobium`, `--include-non-endemic`, and `--no-consolidate`.

4.4 Data QA checks before loading

- Country names in ADMIN0 must exactly match Country values in `df_gdp.csv` after any name mapping.
- Prevalence values should be in percentage points, or the loader should be run in `auto/fraction` mode to convert decimal fractions.
- PopReq and PopTreat should be non-negative and should not be duplicated across species rows unless consolidation is intended.
- Rows with both species should have positive `sm_prev_pct` and `sh_prev_pct`, or species text/flag evidence that the uploader can infer.
- Inspect the country summary from the uploader for unexpected row counts, missing species, and implausible total PopReq.

5. Step-by-step tool workflow

5.1 Sidebar: country, disease, and geography

Control	How to use it	Notes
Select country	Choose the country for the analysis.	The country list comes from <code>df_gdp.csv</code> and must match ESPEN ADMIN0 rows.
Disease module	Choose <i>S. mansoni</i> , <i>S. haematobium</i> , or Both species.	Both species runs both modules and combines programme-level results.
Rows marked both species split	When Both species is selected, set the population share allocated to the mansoni module.	The remaining share is allocated to haematobium. Use this when mixed-species rows do not identify species-specific denominators.
Administrative Unit 1	Select National level or an ADMIN1 unit.	National level uses all rows for the selected country.
Administrative Unit 2	If an ADMIN1 is selected, choose All of ADMIN1 or a specific ADMIN2.	Narrowing geography resets default population and prevalence fields.
Implementing Unit	If an ADMIN2 is selected, choose All of ADMIN2 or a specific IU.	The selected unit label appears in the Country inputs tab.

5.2 Sidebar: MDA programme parameters

Input	Default / range	Use
MDA coverage (%)	Default is PopTreat / PopReq for selected geography; 0-100.	Sets planned population treated and MDA effect multiplier.
MDA frequency	Annual or Biennial.	Annual treats every year. Biennial uses the annual-equivalent effect-vs-annual factor selected in the sidebar and the biennial programme-cost schedule.
MDA target population	SAC only or SAC + at-risk adults.	Controls target multiplier and effective prevalence weighting.
Target population multiplier vs SAC proxy	Default 1.5 when adults included; 1.0 for SAC only.	Multiplies PopReq to approximate total target population.
Discount rate - costs	Default 0.03.	Used for programme cost, health-sector cost, economic impact, and budget impact.

Input	Default / range	Use
Discount rate - effects	Default 0.03.	Used for productivity effects.
Time horizon (years)	5-30, default 10.	Used for discounted productivity and health-sector summaries.
Impact-analysis horizon (years)	3 to time horizon, default up to 5.	Used in Economic impact and Budget impact tabs.
Impact-analysis start year	2020-2050, default 2026.	Calendar year label for impact tables.
Biennial effect vs annual	50-100%, default 70%, shown only for biennial.	Adjusts the MDA effect multiplier under biennial delivery.
PSA iterations	100-10,000, default 1,000.	Higher values improve stable intervals but slow runtime.
Simulation seed	Default 42.	Record this for reproducibility.
Auto-run PSA	Default on.	Turn off when making many input changes.
Run / refresh PSA	Manual button.	Forces PSA refresh after input changes.

5.3 Sidebar: programme costs

Programme cost inputs are editable scenario values. Defaults are placeholders and should be updated with programme budgets or procurement data before policy use.

Cost input	Default	Calculation role
PZQ tablet cost (USD)	\$0.08	Drug cost = tablet cost x tablets/course x planned population treated.
Tablets per treatment course	6	Average tablets per treated person.
Delivery cost per person treated (USD)	\$0.50	Delivery cost = delivery cost x planned population treated.
Mapping / M&E annual cost	\$5,000	Fixed cost in MDA years.
Training cost per MDA round	\$3,000	Fixed cost in MDA years.
Supervision cost per MDA round	\$2,000	Fixed cost in MDA years.
Other annual programme costs	\$1,000	Additional fixed programme cost.

5.4 Sidebar: country economic parameters

These values are populated from df_gdp.csv and can be edited for scenario analysis. The app uses them to calculate daily wage, health-sector unit costs, life expectancy inputs, and cost-effectiveness thresholds.

- Per capita GDP PPP (Int\$): used for GDP-based threshold and the Woods elasticity-based threshold.
- Bottom quintile income share: used to calculate inequality-adjusted daily wage.
- Weekly work hours: used to convert annual income to daily wage.
- Life expectancy: used for years of life lost from fatal bladder cancer.
- OPD and IPD unit costs: inflated to current values using country inflation plus a medical inflation adjustment.

5.5 Sidebar: budget impact resource inputs

Input	Default	Interpretation
Staff minutes per OPD visit averted	15	Used to report staff hours saved from avoided OPD visits.
Staff minutes per IPD bed-day averted	120	Used to report staff hours saved from avoided IPD days.
Staff cost per hour	\$0	Optional monetized staff time value.
Include monetized staff time	Off	Leave off if OPD/IPD unit costs already include staff to avoid double counting.
Additional direct management cost per case averted	\$0 for each morbidity state	Use for diagnostics, medicines, procedures, oncology care, or other costs not captured in OPD/IPD unit costs.

5.6 Disease inputs tab

The Disease inputs tab contains species-specific defaults derived from the selected geography. Always review these values before interpreting outputs.

Module	Key inputs	Interpretation
S. mansoni	Effective prevalence, at-risk population, heavy infection share, hepatomegaly share among heavy infections, hepatic morbidity reduction with MDA, PZQ cure rate.	Defaults use species-specific prevalence and allocated PopReq for the selected geography. Heavy infection drives hepatosplenic morbidity.
S. haematobium	Effective prevalence, at-risk population, female share, background bladder cancer rate, urinary morbidity reduction with MDA, PZQ cure rate, attributable bladder cancer risk reduction, cancer fatality and survival assumptions.	Background bladder cancer rate is interpreted as observed all-cause incidence per 100,000 population. The model estimates the S. haematobium-attributable share using PAF.

5.7 Running and refreshing the PSA

- The app stores simulation outputs in session state and refreshes when core scenario inputs change.
- Use 1,000 PSA iterations for routine analysis; increase only when runtime is acceptable and more stable interval tails are needed.
- Use a fixed seed when results need to be reproducible across runs.
- If making many changes, turn Auto-run PSA off, change inputs, then click Run / refresh PSA once.

6. Interpreting results and downloads

6.1 Country inputs tab

This tab summarizes the selected unit: population requiring MDA, target population used for costing, planned population treated, last recorded MDA coverage, selected MDA coverage, weighted SAC prevalence, weighted adult prevalence, economic parameters, raw ESPEN rows, and programme cost breakdown.

6.2 Results tab

Section	What to review	How to interpret
DALY burden analysis	Morbidity-specific and total DALYs with and without MDA.	DALYs averted are no-MDA DALYs minus MDA DALYs at draw level.
Bladder cancer case impact	All-cause and attributable bladder cancer cases, MDA residual cases, and PAF.	MDA reduces only the attributable component; non-attributable cases remain unchanged.
Time-varying caseload and health sector costs	Projected prevalence, infected counts, and health-sector cost trajectory.	Used to visualize caseload decline and no-MDA vs MDA cost paths.
Productivity losses	Workdays lost with and without MDA, days gained, and productivity gains.	Productivity gains are economic/societal benefits, not budget offsets.
Health sector costs	OPD/IPD-based health-sector costs with and without MDA and annual savings.	Health-sector savings can reduce net incremental programme cost in the cost-effectiveness analysis.
Cost-effectiveness	Combined and species-specific ICERs, CEACs, CE-plane, NMB, and probability cost-effective.	Use the combined programme ICER for the main investment case when both species are selected.

Section	What to review	How to interpret
Return on investment	Return per \$1 invested with PSA interval.	ROI = health-sector savings plus productivity gains divided by programme cost.
Sensitivity analysis	Coverage/frequency scenario comparison.	Use to compare prevalence, cases averted, health-sector savings, and net benefit at years 5 and 10.

6.3 Economic impact tab

Economic impact is a societal/economic perspective. It includes programme costs, health-sector savings, productivity gains, total economic benefits, net economic benefit, cumulative net economic benefit, and benefit-cost ratio over the selected impact horizon. Use this tab for investment-case narratives and societal value discussions.

6.4 Budget impact tab

Budget impact is a health-system/payer perspective. It excludes productivity gains and ROI. It reports gross programme costs, clinical-management budget offsets, staff time saved, optional monetized staff time included in net budget impact, net budget impact, and cumulative net budget impact. Use this tab for programme budget planning and payer affordability discussions.

6.5 Elimination Projections tab

The elimination view projects overall and heavy-intensity prevalence trajectories. EPHP is defined as heavy-intensity infection prevalence in SAC below 1%. The tool also provides an interruption proxy based on near-zero overall prevalence. These projections are scenario tools and should be calibrated with local sentinel-site or surveillance data when available.

6.6 Available CSV downloads

Download file	Created from	Typical use
mansoni_caseloads_with_mda.csv	Results tab, time-varying caseloads	Review projected caseloads under MDA.
mansoni_caseloads_no_mda.csv	Results tab, time-varying caseloads	Comparator for cases averted.
combined_cea_draw_level_outputs.csv	Combined programme ICER section	External CE-plane, NMB, and ICER analysis.
species-specific CEA draw-level CSVs	Species module ICER sections	Module-specific CE review.
mansoni_psa.csv and haematobium_psa.csv	Raw Monte Carlo downloads	Draw-level QA and independent analysis.
combined_daly_psa.csv	Raw Monte Carlo downloads	Combined DALY draw review.
sensitivity_analysis.csv	Sensitivity analysis section	Coverage/frequency scenario comparison.
economic_impact.csv	Economic impact tab	Investment-case and economic impact reporting.
budget_offsets_by_morbidity_state.csv	Budget impact tab	Detailed morbidity-state budget offsets.
budget_impact.csv	Budget impact tab	Programme budget and payer impact reporting.
endgame_scenarios.csv	Elimination Projections tab	Coverage/frequency elimination scenario comparison.

7. Technical assumptions and formulas

7.1 Programme cost formulas

Target population = PopReq x target population multiplier
 Planned population treated = target population x MDA coverage
 PZQ drug cost = tablet cost x tablets per course x planned population treated
 Delivery cost = delivery cost per person x planned population treated

Fixed MDA-year cost = mapping/M&E + training + supervision + other
 MDA-year programme cost = drug + delivery + fixed MDA-year cost
 Off-year fixed cost = fixed MDA-year cost x 0.20
 Annualized cost, annual MDA = MDA-year programme cost
 Annualized cost, biennial MDA = (MDA-year programme cost + off-year fixed cost) / 2
 MDA effect multiplier = MDA coverage fraction x frequency effect factor

7.2 Prevalence and caseload assumptions

- The app treats prevalence as percentage points on a 0-100 scale.
- Effective prevalence is SAC prevalence for SAC-only scenarios. When at-risk adults are included, the tool uses a weighted SAC/adult prevalence based on the target multiplier.
- Caseloads are proportional to at-risk population and effective prevalence.
- Morbidity-state caseloads are derived from infected counts and species-specific morbidity probabilities.
- The Results-tab prevalence trajectory is a deterministic scenario projection for costing and should not be described as a calibrated transmission model or a direct SCHISTOX simulation.

No-MDA comparator: $P_{\text{noMDA}}(t) = P_0 \times (1 + g)^t$. The default annual change g is 0%, so prevalence remains constant unless the analyst changes the sidebar assumption. Non-zero values use ordinary year-on-year compounding.

MDA scenario: $P_{\text{MDA}}(t) = P_{\text{floor}} + (P_0 - P_{\text{floor}}) \times \exp[-\lambda \times I \times t]$.

MDA intensity: $I = \text{annual-equivalent frequency-effect factor} \times \text{selected coverage} / 75\%$. Annual MDA has factor 1.0. Biennial MDA uses the same analyst-selected effect-vs-annual factor used in the PSA and sensitivity analysis; use 50% for a literal every-other-year rounds assumption.

SAC and adult prevalence are projected separately. Adult prevalence decline is applied only when the selected MDA target includes at-risk adults. Species-specific decay defaults are used for *S. mansoni* and *S. haematobium*, with *haematobium* assigned a faster default response; all decay values remain editable scenario assumptions, not fitted transmission parameters.

The residual transmission floor is optional and never forces prevalence upward when the baseline prevalence is already below the floor.

The trajectory section supports caseload and health-sector cost projections. PSA DALYs, ICERs, and CEACs use an annual steady-state effect at the selected coverage and frequency; the deterministic trajectory is a separate time-path preview and should not be used to recalculate the ICER directly. WHO EPHP-style heavy-intensity SAC targets are assessed separately in the Elimination Projections tab.

7.3 DALYs and probabilistic sensitivity analysis

- Each PSA row is one simulation draw.
- Draw-level DALYs averted = no-MDA DALYs - MDA DALYs.
- Reported uncertainty intervals are 95% PSA intervals using the 2.5th and 97.5th percentiles of draw-level outputs.
- Wilson binomial intervals are used for cost-effectiveness probability intervals across PSA draws.
- A fixed seed makes random draws reproducible for a given scenario and iteration count.

7.4 Cost-effectiveness formulas

Incremental cost = annual programme cost - draw-level health-sector savings

ICER = incremental cost / DALYs averted

Net monetary benefit = DALYs averted x willingness-to-pay threshold - incremental cost

Probability cost-effective = share of PSA draws with positive net monetary benefit

ROI = (health-sector savings + productivity gains) / programme cost

If no PSA draws produce positive DALYs averted, the ICER is not evaluable. NMB and probability cost-effective can still be shown across all valid draws.

7.5 Economic impact vs budget impact

Feature	Economic impact	Budget impact
Perspective	Societal/economic	Health-system or payer
Includes programme costs	Yes	Yes
Includes health-sector savings	Yes	Yes, as clinical-management budget offsets
Includes productivity gains	Yes	No
Includes ROI or benefit-cost ratio	Benefit-cost ratio shown	No
Best use	Investment case and societal value	Affordability and budget planning

7.6 Bladder cancer approach

- The bladder-cancer rate input is interpreted as observed all-cause incidence per 100,000 population.
- The *S. haematobium*-attributable fraction is estimated with the Levin population attributable fraction: $PAF = Pe(RR - 1) / [1 + Pe(RR - 1)]$.
- MDA reduces only the attributable bladder-cancer component according to the effective cancer reduction.
- Non-attributable bladder-cancer cases are held constant under MDA.
- Cancer DALYs include years lived with disability and years of life lost using stage-specific fatality, survival duration, mean age, and life expectancy assumptions.

8. Sensitivity and elimination analysis

8.1 Coverage and frequency sensitivity

The sensitivity panel compares coverage and frequency scenarios. It uses a coverage range, selected frequency scenarios, and a biennial annual-equivalent effectiveness factor. Changing the biennial factor changes the biennial prevalence trajectories, cases averted, health-sector savings, net benefit, and plots. Scenario-specific annualized programme costs are applied for annual versus biennial delivery.

Output	Meaning
Prevalence (%)	Projected prevalence at the selected year.
Infected	Projected infected population at the selected year.
Cases averted	Baseline infected population minus scenario infected population.
HS cost discounted	Discounted health-sector cost in the scenario.
HS savings discounted	Discounted savings versus a no-MDA baseline.
Programme cost discounted	Discounted annualized programme cost.
Net benefit	Health-sector savings minus programme cost.

8.2 Elimination projection controls

Control	Use
WHO target year	Year by which to test whether the target is reached, default 2030.
Transmission reservoir floor	Minimum overall prevalence floor; use higher values for more conservative endgame scenarios.
Heavy-intensity clearance acceleration	Multiplier for faster decline in heavy-intensity infection than overall infection.
Target	Choose EPHP heavy SAC <1% or interruption proxy overall ~0%.

Interpret elimination outputs as planning scenarios rather than validated forecasts. The model is intentionally simplified in the low-prevalence tail; local calibration is recommended before making elimination commitments.

9. Quality assurance before production

9.1 Application smoke test

- ☐ App starts with streamlit run app.py.
- ☐ Country list populates from df_gdp.csv.
- ☐ At least one country shows ESPEN rows in the Country inputs tab.
- ☐ National, ADMIN1, ADMIN2, and IU selections update population and prevalence defaults.
- ☐ S. mansoni-only, S. haematobium-only, and Both species scenarios run without errors for appropriate test geographies.
- ☐ The mixed-species population split changes the two module denominators as expected.
- ☐ PSA refresh works with auto-run on and manual refresh.
- ☐ Results, Economic impact, Budget impact, Elimination Projections, and Technical assumptions tabs render.
- ☐ All download buttons produce non-empty CSV files with expected columns.

9.2 Numerical QA checklist

- ☐ Planned population treated = target population x selected coverage.
- ☐ Last recorded MDA coverage = PopTreat / PopReq for the selected geography.
- ☐ Combined programme results are used for main ICER/ROI when both species are selected.
- ☐ ICER uses programme cost minus health-sector savings, not gross programme cost alone.
- ☐ Budget impact excludes productivity gains.
- ☐ Economic impact includes productivity gains.
- ☐ No-MDA values are greater than or equal to MDA values for morbidity states in routine scenarios.
- ☐ Cost-effectiveness NA cases are explained by zero or negative DALYs averted.

9.3 Reproducibility checklist

- ☐ Record app version, dataset version/date, country, geography, disease module, MDA coverage, frequency, target multiplier, discount rates, PSA iterations, and seed.
- ☐ Save exported CSV files used for reporting.

- [] Record any manual overrides to prevalence, population, costs, economic inputs, or cancer parameters.
- [] Store production datasets and results in version-controlled or archived project folders.

10. Troubleshooting

Symptom	Likely cause	Fix
Cannot load required datasets	datasets/consolidated_schisto.csv or datasets/df_gdp.csv is missing, or uploader is not in the expected path.	Add both files under datasets/ and confirm schistosomiasis_data_uploader.py is at the project root.
Country appears in selector but app says no ESPEN rows	Country in df_gdp.csv does not match ADMIN0 in ESPEN data, or rows were excluded as non-endemic.	Align country names or inspect uploader summary and species prevalence columns.
Disease module has zero allocated population	Selected unit has no positive species prevalence for that module, or mixed-species split allocates zero share.	Check sm_prev_pct/sh_prev_pct, species inference, and mixed-species slider.
PSA is slow	High iteration count or auto-run refreshes after each change.	Reduce iterations, turn off auto-run while editing, then run once manually.
Results are blank	Prevalence or at-risk population is zero.	Open Disease inputs and confirm effective prevalence and population are positive.
ICER is NA	No positive DALYs averted in PSA draws.	Check disease prevalence, coverage, MDA effect assumptions, and morbidity reduction parameters.
Budget offset is zero or too small	OPD/IPD utilization or extra management costs are zero, or staff value not included.	Review PSA utilization outputs, additional cost per case fields, and staff-time settings.
Economic and budget impact numbers differ	They are different perspectives.	Use Economic impact for societal value including productivity; use Budget impact for payer affordability excluding productivity.
Elimination target never reached	Coverage/frequency insufficient or reservoir floor prevents crossing threshold.	Review target, coverage, frequency, reservoir floor, and heavy-intensity assumptions.
CSV upload has encoding errors	CSV is not UTF-8 or file pointer was not reset in the upload workflow.	Save as UTF-8 CSV or XLSX, then rerun uploader.

11. Appendices

Appendix A. Default parameter summary

Area	Default values to review
MDA programme	Coverage from PopTreat/PopReg; Annual frequency; SAC + at-risk adults; multiplier 1.5; costs discount 3%; effects discount 3%; time horizon 10 years; impact horizon 5 years; base year 2026; PSA 1,000; seed 42.
Biennial scenarios	Biennial annual-equivalent effect 70% of annual by default; use 50% for a literal every-other-year rounds assumption. Off-year fixed cost equals 20% of fixed MDA-year cost.
Programme costs	PZQ \$0.08/tablet; 6 tablets/course; delivery \$0.50/person; mapping/M&E \$5,000; training \$3,000; supervision \$2,000; other \$1,000.
S. mansoni disease inputs	Heavy infection 40%; heavy infection to hepatomegaly 22%; hepatic morbidity reduction 70%; cure rate 85%.
S. haematobium disease inputs	Female fraction 50%; background bladder cancer 3.5 per 100,000; urinary morbidity reduction 75%; cure rate 87%; attributable cancer reduction 30%; primary CFR 35%; metastatic CFR 85%.
Budget resources	OPD staff 15 minutes; IPD staff 120 minutes; staff hourly cost \$0; staff time value excluded by default; additional management costs \$0 by state.

Appendix B. Glossary

Term	Definition
ADMIN0/1/2	Country and subnational geographic levels from ESPEN-style data.
BIA	Budget impact analysis. Health-system/payer perspective; excludes productivity gains.
CEAC	Cost-effectiveness acceptability curve. Probability cost-effective across willingness-to-pay thresholds.
CET	Cost-effectiveness threshold. The app uses a Woods elasticity-based threshold and also shows 1x GDP.
DALY	Disability-adjusted life year. A measure of health burden combining morbidity and mortality.
EPHP	Elimination as a public health problem. In this tool, heavy-intensity SAC prevalence below 1%.
FGS	Female genital schistosomiasis.
ICER	Incremental cost-effectiveness ratio; incremental cost divided by DALYs averted.
IU	Implementing unit. Lowest geography available in the app selector.
MDA	Mass drug administration.
NMB	Net monetary benefit; DALYs averted times threshold minus incremental cost.
PAF	Population attributable fraction. Used to estimate S. haematobium-attributable bladder cancer.
PopReq	Population requiring MDA.
PopTreat	Population treated in last recorded MDA data.
PSA	Probabilistic sensitivity analysis.
ROI	Return on investment. Benefits divided by programme cost.
SAC	School-age children.

Appendix C. Maintainer notes

- Keep app.py as the Streamlit entry point and schisto_tool/ as the importable package.
- Do not deploy versioned working-copy filenames in production.
- Keep requirements.txt synchronized with runtime imports.
- Run py_compile, smoke_core.py, and regression tests after any logic change.
- When adding new outputs, add a corresponding download button and update this manual.
- When changing cost-effectiveness logic, verify CEAC, CE-plane, NMB, ICER, and draw-level downloads together.
- When changing budget impact logic, confirm productivity gains remain excluded from budget impact unless intentionally redesigning the perspective.